GEYI YANG

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Shenzhen, Guangdong, China

Research Interests

LLM-based agents, GUI agents, self-improving agentic systems, harness optimization, and retrieval-augmented generation (RAG).

Education

- **The Chinese University of Hong Kong, Shenzhen** *Sep. 2023 – Present*
  B.Eng. in Computer Engineering Shenzhen, China
  - GPA: 3.73/4.00.
  - Selected coursework: Artificial Intelligence, Machine Learning, Optimization, Data Structures, Operating Systems, and Internet Architecture and Protocol.
- **University of California, Berkeley** *Jan. 2026 – Jun. 2026*
  GLOBE Program Visiting Student in Computer Science Berkeley, CA, USA

Honors and Awards

Academic Performance Scholarship Awardee, CUHK-Shenzhen (top 8%) 2023–2024
Dean's List, CUHK-Shenzhen 2023–2024, 2024–2025, 2025–2026
COMAP Mathematical Contest in Modeling (MCM), Finalist (F Award; top 1%) 2026
China International College Students' Innovation Competition (CICSIC), Bronze Award 2025

Publication

[1] **Geyi Yang**, Zikun Qu, Xiang Li, Zhiyong Wang, Min Zhang, Shipei Zeng, and Zhongxiang Dai. "GUI-HARVEST: Self-Improving GUI Agents through Evidence-Driven Harness Evolution." *arXiv preprint arXiv:2610.00948*, 2026. Under review at ICLR 2027 (scores: 4/3/3/2, out of 4; confidence: 3/3/3/2). [arXiv]

Research Experience

- **GUI-HARVEST: Self-Improving GUI Agents through Evidence-Driven Harness Evolution** *Jun. 2026 – Sep. 2026*
  Research Assistant, Shenzhen Research Institute of Big Data | *Advisor: Zhongxiang Dai* GitHub
  - First-authored GUI-HARVEST (under review at ICLR 2027), an automatic harness optimizer that enables self-improving GUI agents with frozen backbone models by converting repeated multimodal executions into validated runtime code changes.
  - Designed an evidence-driven loop that aligns actions with before-and-after screenshots, compares repeated runs, clusters recurring cross-task failures, and promotes edits only after score and predicted-behavior checks.
  - Achieved held-out gains across six open and proprietary backbones on OSWorld-Verified; Qwen3-VL-32B-Instruct gained 12.33 points on the full suite, while frozen-harness transfer improved GPT-5 by 13.87 points on WindowsAgentArena.
- **Privacy-Aware Disclosure in Multi-Step LLM Workflows** *Sep. 2026 – Present*
  Capstone Research | *Advisor: Jianwei Huang*
  - Built a testbed for privacy-aware LLM agents, auditing benchmark tasks and constructing a two-step banking scenario to compare early versus on-demand disclosure under mutual-information budgets; kept private state and utility scoring local.
  - Implemented tool execution and utility scoring; evaluated finite disclosure policies across four privacy budgets using 120 cloud-LLM workflows, and compared 24 baseline configurations to characterize disclosure timing and privacy–utility trade-offs.
- **Intrinsic Signals and Answer Diversity for LLM Reasoning** *Sep. 2025 – Dec. 2025*
  Independent Study | *Advisor: Yixiang Fang*
  - Explored label-free RL rewards using attention-based token dependency graphs; evaluated structural metrics against reasoning correctness and identified response length and problem difficulty as confounders.
  - Analyzed Qwen 3B/7B on MATH and AIME (4 seeds; up to 4,096 rollouts/problem), finding answer-frequency and entropy convergence in 92% of model–problem pairs despite new answers emerging. Proposed a lightweight pre-sampling predictor of entropy and effective answer counts to guide adaptive computation and test-time RL rewards.

Internships

- **Dongguan Zhongke Institute of Cloud Computing** *Mar. 2025 – Jun. 2025*
  Domain-Specific Agentic RAG for Mining-Blasting QA Dongguan, China
  - Built an end-to-end Agentic RAG system for mining-blasting QA using LangGraph memory, query planning, answer synthesis, and ReAct tool use, with human clarification for ambiguous queries.
  - Developed Docling-based parsing and parent-child hybrid retrieval in Qdrant, combining BGE-M3, BM25, and BGE-Reranker-v2-m3; achieved 81% Recall@5 and 78% answer accuracy on 300+ QA cases.
- **Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** *Sep. 2023 – Jun. 2024*
  Multimodal Digital Human Synthesis for Emotional Interaction Shenzhen, China
  - Developed an end-to-end "Digital Relative" pipeline from family photos and recordings, integrating MockingBird voice cloning, SadTalker lip-sync and facial animation, and ChatGPT-powered dialogue.
  - Piloted the system with family users in 500+ interactions: 83% reported a familiar emotional connection and 20% reported a strong emotional response.

Products

- **StashDock: Your Context Library for Any AI** [GitHub stars / Product Hunt rank: TBD]
  Creator and Full-Stack Developer | *Oct. 2026 – Present* Website | GitHub
  - Designed and launched an AI-native, visualized personal context library for managing and combining links, files, and prompts into reusable context sets. Enabled one-click addition to active AI conversations and AI-assisted reference discovery, filtering, and saving.
  - Engineered Windows/macOS apps with Tauri/Rust, React/TypeScript, and SQLite; integrated a Chrome capture extension and a ChatGPT plugin in the conversation sidebar, with cross-app clipboard/drag-and-drop transfer and MCP tools/Skills for AI-assisted context management.

Selected Projects

- **MCM 2026: Dynamic Hybrid Lunar Logistics Optimization (Finalist, top 1%)** *2026*
  Modeling and Programming Lead Article
  - Developed a dynamic Earth-to-Moon logistics model integrating space elevators, reusable rockets, and ISRU maturation; formulated time-cost objectives and operational constraints.
  - Used Differential Evolution (1,301 evaluations), Monte Carlo analysis (5,000 simulations), and sensitivity analysis to identify a robust three-phase transition strategy; awarded Finalist (top 1%).
- **Meme-Qwen (7B-Instruct)** *Jan. 2026*
  Chinese Internet-Meme Conversational Language Model Model | Dataset
  - Independently built and open-sourced, to our knowledge, the first Chinese internet-meme conversational SFT dataset and the first open model of this type: collected 30,000+ post-comment pairs with Playwright and released 8,680 ChatML examples.
  - Fine-tuned Qwen2.5-7B on a single A100 (40GB) with LoRA and DPO; improved meme-use accuracy by 32%, with human and LLM preference rates of 82%/91%/68% for humor, accuracy, and naturalness.
- **Telos: AI-Generated, Ready-to-Learn Courses** *Aug. 2026 – Present*
  Independent Full-Stack Developer GitHub
  - Building a goal-conditioned education product that turns user materials and retrieved resources into ready-to-learn, multimodal courses; designed an agentic pipeline for deduplication, quality assessment, sequencing, and embedding source text, images, and videos in one learning workspace.
- **Voice Turing Test: WeChat Mini Program and Research Data Platform** *May 2025 – Aug. 2025*
  Full-Stack Development Lead GitHub
  - Independently led registration, UI/UX, full-stack development, testing, and deployment of a gamified Human-vs-AI speech challenge used as the data-collection platform for *Human or Machine?* (ICLR 2026).
  - Built the native WeChat front end and Tencent Cloud backend with bilingual UI, real-time scoring, achievements, leaderboards, and referral rewards; reached 1,000+ users, 300+ DAU, and 5,000+ valid records in the first month.

Teaching Experience

- **The Chinese University of Hong Kong, Shenzhen** *Sep. 2025 – Dec. 2025*
  Teaching Assistant – MAT1001 Calculus I Shenzhen, China
  - Served as a teaching assistant for the first-year calculus course MAT1001, supporting tutorials and student learning.

Skills and Languages

Programming & Systems: Python, C++, JavaScript, TypeScript, Shell, FastAPI, React, SQLite, Git, Linux.
Machine Learning & Agents: PyTorch, Transformers, LoRA/PEFT, TRL (DPO), RAG, LangChain, LangGraph, GUI agents, Qdrant, MCP.
Languages: English (TOEFL 111; GRE 332).